Non-Independence in Nested Designs

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Non-independence in a nested example.

	Instructions (j)					
	Competition			No Competition		
	Group 1	Group 2	Group 3	Group 4	Group 5	Group 6
Y_{hij}	8	6	7	6	7	5
Y_{hij}	9	7	8	6	6	5
Y_{hij}	7	6	8	5	7	4
Group Mean	8	6.33	7.67	5.67	6.67	4.67
Condition Means	7.33			5.67		
Grand Mean	6.50					

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Grand Mean	6.50					

Source	SS	df
Condition	12.50	1
Error Between Groups	10.67	4
Total Between Groups	23.17	5
Total Within Group	5.33	12
Total	28.50	17

Remember, h = *individual*, i = *group*.

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$$\sum \left(Y_{hij} - \bar{\bar{Y}}\right)^2$$

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$$\sum\left(Y_{hij} - \bar{Y}_i\right)^2$$

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$$\sum (\bar{Y}_i - \bar{\bar{Y}})^2 \, n_i$$

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$$\sum \left(\bar{Y}_i - \bar{\bar{Y}}\right)^2 n_i$$

NOTE THAT I ACCIDENTALLY LEFT "n" OUT OF THE SLIDES IN THE VIDEO.

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$$\sum\left(\hat{Y}_{hij} - Y_{hij}\right)^2 - \boxed{\sum\left(Y_{hij} - \bar{Y}_i\right)^2}$$

Our classic SSE(A), but we have to take SSwithin out of the calculation.

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$$\sum\left(\hat{Y}_{hij} - \bar{\bar{Y}}\right)^2$$

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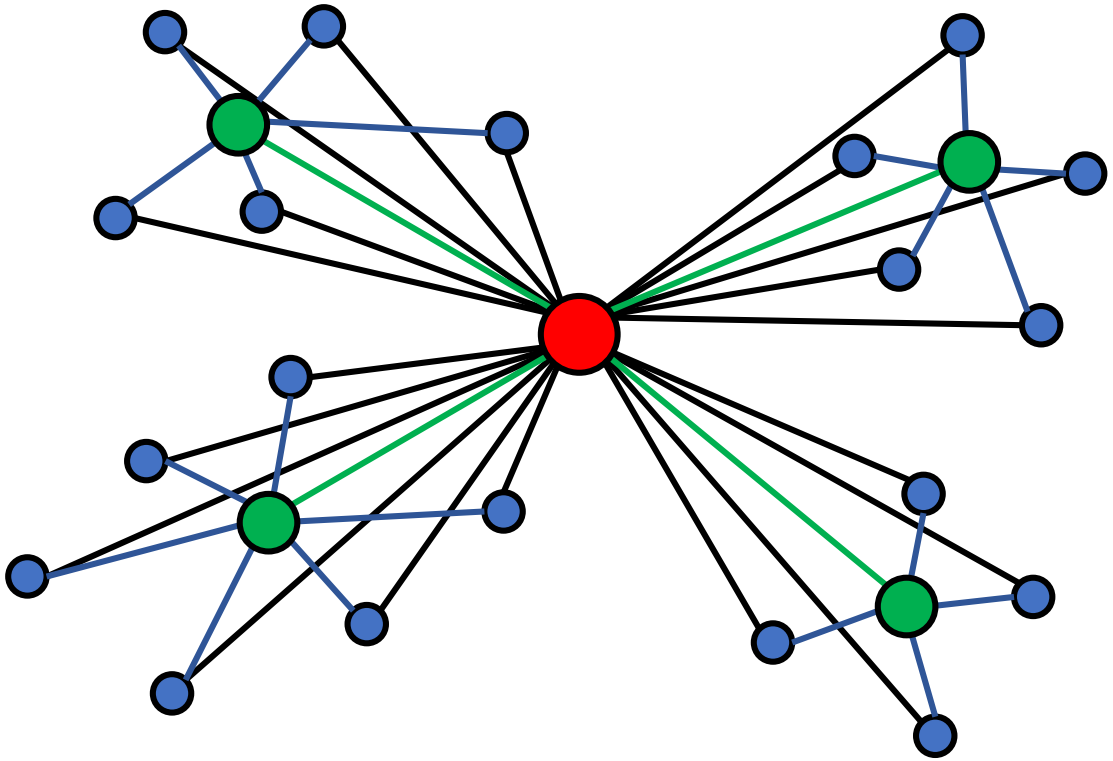
Our calculation of the MSR and MSE, and subsequently the F_observed, now relies on this SSE(A) with the within group variation removed.

The magic of squares!



$$SS_{\text{total}} = \Sigma(X - \overline{M})^2$$

$$SS_{\text{within}} = \Sigma_i(\Sigma(X_i - M_i)^2)$$

$$SS_{\text{effect}} = \Sigma_i(n_i(M_i - \overline{M})^2)$$